Peripheral T-cell lymphoma

Nodal PTCL · ALCL · NK/T · ATLL · rare extranodal subtypes — **CD30 and ALK status pick the induction; ALK+ ALCL is the only subtype CHOP-like therapy reliably cures**

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin, T-cell lymphoma · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

ENTITIES — WHO-HAEM5 / ICC 2022

Entity	What defines it
PTCL-NOS *~26%*	Diagnosis of exclusion; CD4 > CD8, loss of CD5/CD7 common. GATA3-high subset does worse than TBX21
Nodal TFH lymphoma, angioimmunoblastic type *~19%*	**Formerly AITL** (ICC: follicular helper T-cell lymphoma). Older adults; rash, effusions, **polyclonal hypergammaglobulinemia, Coombs+ hemolysis**, EBV+ B-blasts. PD-1, CXCL13, ICOS, BCL6, CD10. **TET2, DNMT3A, IDH2 R172, RHOA G17V**. Follicular and NOS types share the TFH phenotype
ALCL, ALK+	Younger; NPM1::ALK t(2;5); strong uniform CD30. **5-yr OS ~85%** — the good one
ALCL, ALK–	Test FISH: **DUSP22-R (~30%)** 5-yr OS 90% · **TP63-R (~8%)** 17% · neither 42%
Breast implant-associated ALCL	**Textured implants**; late seroma or capsular mass, CD30+ ALK–. Tap the seroma → cytology + CD30
Extranodal NK/T-cell *~10%*	WHO-HAEM5 drops "nasal type." **EBV+ always**, CD56+, cCD3ε+, surface CD3–; angiocentric, necrotic mid-face mass; Asia and Latin America
ATLL *~10%*	**HTLV-1**; Japan, Caribbean. CD4+CD25+ "flower cells," **hypercalcemia**, lytic lesions, skin
EATL · MEITL	EATL: celiac (refractory sprue). **MEITL** (old "type II EATL"): no celiac, CD8+CD56+, **SETD2** loss. Perforation risk
Hepatosplenic	γδ TCR; young men, **IBD on anti-TNF + thiopurine**; hepatosplenomegaly, sinusoidal marrow, no nodes. Dismal

Frequencies from the International T-cell Project (Vose, JCO 2008, n=1,153) — **10% of referral diagnoses were misclassified**, so insist on expert hematopathology review.

TREATMENT PATHWAY

Nodal PTCL — frontline

Induction by CD30 and subtype
CD30 ≥10%, esp. sALCL: BV-CHP ×6 (ECHELON-2)
CD30–: CHOP or **CHOEP** (age ≤60) ×6
EATL: CHOP ×1 → IVE/MTX → auto-HCT

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Consolidate in CR1?
ALK+ ALCL and DUSP22-R: no — observe
others, fit, in CR: **consider auto-HCT**
evidence is **non-randomized** and conflicting

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Limited stage
CHOP-based ×3–4 + ISRT, or ×6 ± RT

Relapsed / refractory

Transplant candidate
salvage (ICE, GDP, DHAP) or single agent → **allo-HCT** (auto if chemosensitive, no prior ASCT)

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Match the agent to the biology
CD30+: BV (sALCL ORR 86%, CR 57%)
ALK+: crizotinib or alectinib, brigatinib, lorlatinib
TFH/AITL: duvelisib, HDAC inhibitor, aza
any: belinostat, pralatrexate, gemcitabine, benda

Extranodal NK/T-cell

Localized (IE–IIE)
RT ≥50 Gy + asparaginase-based chemo
concurrent DeVIC or sequential P-GemOx / DDGP
no anthracyclines — P-glycoprotein resists them

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Advanced or relapsed
DDGP or mod-SMILE → **allo-HCT** if fit
R/R after asparaginase: **PD-1 blockade**
follow **plasma EBV DNA** as the tumor marker

ATLL (HTLV-1)

By Shimoyama subtype
smoldering / favorable chronic: **watch** or AZT/IFN
acute: **AZT + IFN-α** or chemo, then **allo-HCT**
lymphoma type: **chemo** (CHOP, mLSG15) → allo
mogamulizumab: R/R (Japan) — GVHD risk pre-allo

WORKUP

- **Excisional or core biopsy** with T-cell panel, **CD30** (% of cells), ALK, TFH markers, EBER; **TCR clonality**. ALK– ALCL → **DUSP22 and TP63 FISH**
- **PET/CT**, marrow biopsy, LDH, **HTLV-1 serology** (and calcium) in anyone at risk; **plasma EBV DNA** in NK/T and AITL
- HBV/HCV/HIV; echo before anthracycline. **Nasal NK/T: MRI of the nasopharynx** + ENT exam
- **ATLL: screen for Strongyloides** (co-endemic with HTLV-1) before steroids

PROGNOSIS — PIT AND IPI

PIT factors	5-yr OS	10-yr OS
0	62%	55%
1	53%	39%
2	33%	18%
3–4	**18%**	**13%**

PIT (Gallamini, Blood 2004): **age >60, LDH >ULN, ECOG ≥2, marrow involvement**. The **IPI** also stratifies PTCL and is the stratifier in ECHELON-2. **NK/T uses PINK** (age >60, stage III–IV, distant nodes, non-nasal primary; 3-yr OS 81/62/25% for 0/1/≥2); **PINK-E** adds detectable EBV DNA.

ATLL AND RARE SUBTYPES

- **ATLL subtypes (Shimoyama):** acute, lymphoma, chronic (unfavorable if LDH, BUN or albumin abnormal), smoldering. **AZT + IFN-α** (meta-analysis, n=254): 5-yr OS 46% with first-line antivirals — benefit in **acute, chronic and smoldering**, while **lymphoma type does better with chemo**
- **mLSG15** (VCAP-AMP-VECP) beat biweekly CHOP on CR (40% vs 25%), 3-yr OS 24% vs 13% (NS). **Mogamulizumab**: R/R ATLL ORR 50% (Japan-approved, not US). **Given <50 d before allo-HCT it raises severe GVHD and NRM** (1-yr OS 32% vs 49%)
- **BIA-ALCL:** disease confined to capsule/seroma → **total en bloc capsulectomy + implant removal** (both sides) is usually curative; nodal or unresectable → BV-CHP
- **Hepatosplenic:** CHOP fails — ICE or IVAC-type induction, then **allo-HCT in CR1**
- **EATL/MEITL:** nutrition + surgical review for perforation; IVE/MTX (Newcastle) → auto-HCT. **AITL:** steroids/cyclosporine for immune flares; watch for concurrent EBV+ B-cell lymphoma

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FRONTLINE EVIDENCE

- **ECHELON-2 (BV-CHP vs CHOP, CD30 ≥10%):** PFS HR 0.71, **OS HR 0.66** — the only regimen to beat CHOP on survival. 5-yr PFS **51% vs 43%**, OS **70% vs 61%**
- **But the benefit sits in systemic ALCL** (70% of enrollment; 5-yr PFS HR 0.55). PTCL-NOS HR 0.75 (NS); **AITL favored CHOP** (small n). The label is **sALCL or other CD30-expressing PTCL**; outside ALCL it is a reasonable, not proven, choice
- **CD30 threshold was ≥10%** by local IHC; retreatment with BV at relapse after BV-CHP still gave ORR 59%
- **CHOEP** (DSHNHL pooled analysis): etoposide improved 3-yr EFS **75% vs 51%** in age ≤60 with normal LDH; **over 60, CHOP-21 stays standard**
- **Auto-HCT in CR1 — never randomized.** NLG-T-01 (CHOEP-14 → ASCT, ALK+ excluded): 5-yr OS 51%, PFS 44%, TRM 4%. COMPLETE cohort: ASCT in CR1 OS HR 0.37, driven by AITL; a LYSA registry analysis found no benefit. Offer it to fit non-ALK+ patients in CR, especially AITL and higher IPI

EXTRANODAL NK/T-CELL LYMPHOMA

- **Anthracyclines fail** — tumor cells express P-glycoprotein (MDR1). Build regimens on **L-asparaginase/pegaspargase**, methotrexate, gemcitabine, platinum, etoposide
- **Localized:** radiation is the backbone. **DeVIC ×3 + concurrent RT 50 Gy** (JCOG0211): 2-yr OS **78%** vs 45% historical RT alone. Sequential/sandwich P-GemOx or DDGP + RT is the common alternative
- **Advanced: DDGP beat SMILE** in a randomized trial (n=87; PFS HR 0.42, OS HR 0.41, 5-yr OS 74% vs 52%, less myelosuppression). SMILE: ORR 79% but **grade 4 neutropenia 92%** — require ALC ≥500
- **EBV DNA** in plasma: detectable at diagnosis is adverse (PINK-E); **persistent after therapy predicts relapse**
- **R/R after asparaginase: PD-1 blockade** — pembrolizumab responded in 7/7 (Kwong, Blood 2017); off-label in the US. Consolidate responders with allo-HCT

RELAPSED / REFRACTORY AGENTS — US STATUS

Agent	Evidence and status (Sept 2026)
Brentuximab vedotin *anti-CD30 ADC*	R/R sALCL ORR **86%**, CR 57% (Pro, JCO 2012). Regular approval. Activity in CD30+ non-ALCL is lower and not CD30-level dependent
Pralatrexate *antifolate*	PROPEL: ORR **29%**, DOR 10.1 mo, PFS 3.5 mo. **Accelerated approval 2009 — still unconfirmed**; the confirmatory trial will not read out before ~2030
Belinostat *pan-HDAC*	BELIEF: ORR **26%**, CR 11%, DOR 13.6 mo, but PFS 1.6 mo. **Accelerated 2014, unconfirmed** — ODAC (Nov 2023) criticized both delays. Least myelosuppressive; OK with low platelets
Romidepsin *HDAC 1/2*	ORR 25%, DOR 17 mo. **PTCL indication withdrawn 2021** (Ro-CHOP negative) — still NCCN-listed and still FDA-approved for **CTCL**
Duvelisib *PI3K-δ/γ*	PRIMO (JCO 2026, n=123): ORR **48%**, CR 33%, but PFS 3.4 mo; **AITL ORR 62%, CR 51%**, PFS 8.3 mo. **Not FDA-approved for PTCL** (phase 3 TERZO ongoing)
Valemetostat *EZH1/2*	VALENTINE-PTCL01: ORR **44%** (52/119). **Approved in Japan** (ATLL 2022, PTCL June 2024); **not US-approved**
Golidocitinib *JAK1*	JACKPOT8 B: ORR **44%**, CR 24% (94% Asian). **Conditional approval in China** (June 2024); US fast-track only
ALK inhibitors	**Crizotinib FDA-approved Jan 2021 for age 1–21** (ORR 88%); adults off-label. Alectinib, brigatinib, lorlatinib also active
Others	Gemcitabine, bendamustine (ORR 50%, PFS 3.6 mo), lenalidomide (AITL), **azacitidine in TET2-mutant TFH**, cyclosporine for AITL immune phenomena

Response rates are all single-arm and PFS is short — use these to bridge a responder to **allo-HCT**, the only strategy with a durable plateau in R/R PTCL.

TOXICITY AND MONITORING

- **Brentuximab: peripheral neuropathy** — hold for grade 3, dose-reduce for grade 2. **PML** (boxed warning); pancreatitis. **Never with bleomycin** (pulmonary toxicity)
- **Pralatrexate: mucositis** is dose-limiting and prevented by B12/folate; check CBC weekly; reduce for CrCl <30
- **HDAC inhibitors:** thrombocytopenia, nausea, fatigue, **QTc** — ECG, keep K and Mg replete. **Belinostat: UGT1A1 genotype** guides the dose
- **Duvelisib boxed warnings:** fatal infections, **colitis**/**diarrhea**, severe skin reactions, pneumonitis; transaminitis. Mandatory PJP prophylaxis; CMV monitoring
- **Asparaginase:** hypersensitivity, **pancreatitis**, hepatotoxicity, **hypofibrinogenemia and thrombosis**, hyperglycemia. Check fibrinogen, lipase, LFTs, glucose each cycle
- **HD-MTX (SMILE):** hydration + **urine alkalinization (pH ≥7)**, MTX levels, leucovorin rescue; **glucarpidase** if delayed clearance with AKI; stop PPIs, TMP-SMX, NSAIDs around the dose
- **Ifosfamide:** mesna; watch for encephalopathy (methylene blue) and Fanconi-type nephrotoxicity
- **ALK inhibitors:** crizotinib — visual disturbance, transaminitis, QTc, bradycardia
- Prophylaxis: **acyclovir + TMP-SMX** with HDAC, duvelisib, SMILE, alemtuzumab; **entecavir if HBcAb+**

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REGIMEN DOSING

Regimen	Dosing
BV-CHP *ECHELON-2*	Brentuximab **1.8 mg/kg** (max 180 mg) + cyclophosphamide 750 mg/m² + doxorubicin 50 mg/m² d1, prednisone 100 mg d1–5 — **q21d ×6–8. No vincristine.** Primary G-CSF
CHOP / **CHOEP**	CHOP as above + vincristine 1.4 mg/m² (max 2) d1; CHOEP adds **etoposide 100 mg/m² d1–3** — q21d ×6 (NLG used CHOEP-14)
Brentuximab (R/R)	1.8 mg/kg IV q21d up to 16 cycles; 1.2 mg/kg if Child-Pugh A or neuropathy grade 2
Pralatrexate	**30 mg/m²** IV push weekly ×6 q7wk. **Folic acid 1–1.25 mg daily** from 7–10 d before until 30 d after; **B12 1 mg IM q8–10 wk**. Leucovorin mouthwash for mucositis
Belinostat	**1,000 mg/m²** IV over 30 min d1–5 q21d; **750 mg/m² if UGT1A1*28 homozygous**
Romidepsin	14 mg/m² IV over 4 h d1, 8, 15 q28d — K ≥4, Mg ≥2 before each dose; QTc
Duvelisib *off-label*	**75 mg PO BID ×2 cycles, then 25 mg BID**, q28d continuous. PJP + HSV/VZV prophylaxis
Gemcitabine	1,200 mg/m² d1, 8, 15 q28d ×3–6
DeVIC + RT *2/3 dose*	Dexamethasone 40 mg d1–3 + ifosfamide 1,000 mg/m² d1–3 (+ mesna) + etoposide 67 mg/m² d1–3 + carboplatin 200 mg/m² d1 — q21d ×3, **concurrent RT 50 Gy**
DDGP	Dexamethasone 15 mg/m² d1–5 + cisplatin 20 mg/m² d1–4 + gemcitabine 800 mg/m² d1, 8 + **pegaspargase 2,500 IU/m² d1** — q21d ×6
Modified SMILE	MTX 2 g/m² d1 (leucovorin from 24 h) · ifosfamide 1.5 g/m² + mesna, dexamethasone 40 mg, etoposide 100 mg/m² d2–4 · **pegaspargase 2,000 U/m² d7** once MTX <0.04 µM — q28d ×2–6. **Urine alkalinization for HD-MTX**
AZT + IFN *ATLL*	High-dose oral zidovudine + daily SC interferon-α, continued indefinitely while responding — **no randomized dosing standard**; follow the treating center's protocol

DDGP doses are from the Chinese trial protocol (Li, Clin Cancer Res 2016). Interferon alfa-2b (Intron A) is off the US market — pegylated IFN is the practical substitute.

KEY TRIALS

Trial	Design	Result and meaning
ECHELON-2 *Lancet 2019*	Phase 3, double-blind, n=452, CD30 ≥10%, 70% sALCL	BV-CHP vs CHOP: mPFS **48.2 vs 20.8 mo**, HR 0.71; **OS HR 0.66**; 5-yr OS 70% vs 61%. Benefit concentrated in sALCL
NLG-T-01 *JCO 2012*	Single-arm phase 2, n=166, ALK+ excluded	CHOEP-14 → ASCT: 72% reached transplant; **5-yr PFS 44%**, OS 51%, TRM 4%
PROPEL *JCO 2011*	Single-arm, n=115	Pralatrexate: ORR 29%, DOR 10.1 mo, PFS 3.5 mo — accelerated approval, unconfirmed
BELIEF *JCO 2015*	Single-arm, n=129	Belinostat: ORR 26%, CR 11%, DOR 13.6 mo, PFS 1.6 mo
PRIMO *JCO 2026*	Single-arm, n=123	Duvelisib: ORR 48%, CR 33%; **AITL ORR 62%, CR 51%**. Not FDA-approved for PTCL
VALENTINE-PTCL01	Single-arm, n=133	Valemetostat: ORR 44%. Approved in Japan only
JACKPOT8 B	Single-arm, n=104, 94% Asian	Golidocitinib: ORR 44%, CR 24%. China only
JCOG0211 *JCO 2009*	Phase 1/2, n=33, stage IE–IIE nasal	DeVIC + RT 50 Gy: **2-yr OS 78%** vs 45% historical RT alone
DDGP vs SMILE *JAMA Oncol 2022*	Randomized, n=87, stage III–IV NK/T	PFS HR 0.42, OS HR 0.41; 5-yr OS 74% vs 52%; less myelosuppression
JCOG9801 *JCO 2007*	Phase 3, n=118, aggressive ATLL	mLSG15 vs CHOP-14: CR 40% vs 25%; 3-yr OS 24% vs 13% (NS)
AZT/IFN meta-analysis *JCO 2010*	n=254, retrospective	First-line antivirals: 5-yr OS 46%; benefit in leukemic subtypes, not lymphoma type